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Financial Analytics Lab

GROUP 11

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Answers

[Link to PYTHON CODE](#)

Q What is VAR model?

VAR models (vector autoregressive models) are used for multivariate time series. The structure is that each variable is a linear function of past lags of itself and past lags of the other variables. In general, for a VAR(p) model, the first p lags of each variable in the system would be used as regression predictors for each variable.
The VAR model is widely used in econometrics for forecasting and policy analysis, particularly in macroeconomics where multiple economic variables are interrelate. It is an extension of univariate autoregressive models to multivariate time series data.
A pth-order VAR model is written as

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + e_t$$

Q What is Granger Causality Test?

The Granger causality test is a statistical hypothesis test used to determine whether one time series can be used to forecast another. In other words, it helps determine if past values of one variable (X) contain information that can improve forecasts of another variable (Y), above and beyond the information provided by past values of Y alone. The test doesn't necessarily imply true cause-and-effect, but rather a predictive relationship. After ensuring stationarity of the variables, lag lengths are chosen, and regression models are fitted with and without lagged values of the potential causal variable. A significance test, often an F-test, determines if including the lagged values improves prediction. If so, it suggests Granger causality. However, it doesn't confirm deterministic causation and assumes no omitted variables.

Answers

Q What are the limitations of VAR (p) model?

Vector Autoregression (VAR) models are widely used in econometrics and time series analysis for modeling the dynamic relationships among multiple variables. However, they do have several limitations:

- **Stationarity Assumption**: VAR models typically assume that the time series variables are stationary. If the variables are non-stationary, spurious relationships may be detected, leading to unreliable results.
- **Curse of Dimensionality**: VAR models become computationally intensive as the number of variables increases.
- **Parameter Estimation**: Estimating VAR parameters requires a relatively large number of observations, especially when the number of variables and lag orders are high.
- **Model Specification**: Choosing the appropriate lag order (p) in a VAR model is crucial. Selecting too few lags may result in omitted variable bias, while selecting too many lags may lead to overfitting and poor out-of-sample forecasting performance.
- **Dynamic Effects**: VAR models assume that the relationship between variables is constant over time. However, in reality, these relationships may change over different time periods
- **Endogeneity**: VAR models assume that the variables are exogenous. In cases where variables are endogenous, such as in simultaneous equation models, failing to account for endogeneity may lead to biased and inconsistent parameter estimates.
- **Forecasting Performance**: While VAR models are commonly used for forecasting, their performance may deteriorate when the underlying data generating process is nonlinear or exhibits structural breaks.

Answers

Q .“If the primary object is forecasting, VAR will do the job.” Critically evaluate this statement.

The statement "If the primary object is forecasting, VAR will do the job" suggests that Vector Autoregression (VAR) models are particularly well-suited for forecasting purposes. Let's critically evaluate this statement:
Strengths of VAR for Forecasting:

- **Simultaneous Equation Modeling:** VAR allows for the simultaneous estimation of multiple interrelated variables, making it suitable for capturing complex interactions among variables.
- **Flexibility:** VAR models do not require strict assumptions about the underlying data distribution or relationships among variables, making them versatile for various types of data.
- **Dynamic Nature:** VAR models capture the dynamic relationships between variables over time, allowing for the incorporation of time-series information into forecasts.
- **Multiple Interrelated Variables**: VAR excels when dealing with multiple time series data that influence each other. It captures these interdependencies, leading to potentially more accurate forecasts compared to univariate models.

Alternatives for Forecasting:

- **Univariate Models (ARIMA):** When dealing with a single time series, simpler models like ARIMA can be more efficient and interpretable.
- **Machine Learning Techniques:** For complex relationships or non-linear dependencies, machine learning algorithms like Random Forests or Gradient Boosting can outperform VAR.

In conclusion, VAR can be a powerful tool for forecasting with multiple interrelated time series, but its limitations like overfitting and interpretability should be considered. The choice of forecasting model depends on the specific data characteristics and forecasting goals.

Answers

Step 1: Stationarity Check (ADF test):

```
GDP Growth variable is not stationary
Stock Index variable is stationary
Repo Rate variable is stationary
Unemployment Rate variable is stationary
```

Step 2: Since GDP Growth came out to be non stationary, we use differencing method to induce stationarity

```
df['GDP Growth'] = df['GDP Growth'].diff()
df['GDP Growth'].dropna()
```

```
Date
1999-06-30    2.6
1999-09-30    0.5
1999-12-31    1.3
2000-03-31   -0.8
2000-06-30    0.3
             ...
2020-09-30    0.7
2020-12-31    0.3
2021-03-31   -0.2
2021-06-30   -0.1
2021-09-30   -0.4
Name: GDP Growth, Length: 90, dtype: float64
```

```
[17] df = df.replace([np.inf, -np.inf], np.nan).dropna()
```

```
[18] result = adfuller(df['GDP Growth'])
     print(f'p-value:{result[1]}')
```

```
p-value:9.589125356585001e-09
```

Here, p-value came out to be less than 0.05 after using differencing method

Answers

Step 3: We find optimal lag length(optimal lag length = 9)

```
Lag Order: 1, AIC: 11.884305603326782, BIC: 12.443549731356477
Lag Order: 2, AIC: 12.095734237665658, BIC: 13.109190207224925
Lag Order: 3, AIC: 12.163942875903333, BIC: 13.637818992800327
Lag Order: 4, AIC: 12.046962317681748, BIC: 13.987609017044988
Lag Order: 5, AIC: 12.051664414260678, BIC: 14.465578597145225
Lag Order: 6, AIC: 11.96886380514493, BIC: 14.862693327577444
Lag Order: 7, AIC: 11.54435443847511, BIC: 14.924902757805295
Lag Order: 8, AIC: 11.479582035916248, BIC: 15.353813019317244
Lag Order: 9, AIC: 11.327233899823995, BIC: 15.702276799719318
Lag Order: 10, AIC: 11.600274765570731, BIC: 16.483429366652185
Lag Order: 11, AIC: 11.66167984578115, BIC: 17.06042178811107
Lag Order: 12, AIC: 11.441340068003566, BIC: 17.363326350454333
Best Lag Order based on AIC: 9
```

Step 4: We fit our data on VAR model using optimal lag length

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:             Thu, 04, Apr, 2024
Time:                     10:16:40
--------------------------------------------------------------------
No. of Equations:         4.00000    BIC:                    15.7023
Nobs:                     81.0000    HQIC:                   13.0826
Log likelihood:          -770.489    FPE:                    111179.
AIC:                      11.3272    Det(Omega_mle):         24685.0
--------------------------------------------------------------------
```

```
Coefficients for Lag Order L9:
[[-1.85185134e-01  5.19162997e-04 -3.40001041e-01  6.01444764e-02]
 [-5.54206654e+01 -9.73270630e-02 -9.23813777e+01  3.95287446e+01]
 [ 2.87241534e-02  1.42305864e-04 -5.10287148e-02  1.24148547e-01]
 [ 2.57025038e-01  2.17015553e-04  4.24606647e-02 -1.90780308e-01]]
```

Answers

Step 5: Stability Criteria

```
coefs = model_fit.coefs

# Check stability condition for each coefficient matrix
for i, coef_matrix in enumerate(coefs):
    eigenvalues = np.linalg.eigvals(coef_matrix)
    if i == 8 and all(abs(eig) < 1 for eig in eigenvalues):
        print(f"Coefficient matrix (Lag : {i+1}) is stable.")
    elif i==8:
        print(f"Coefficient matrix (Lag : {i+1}) is unstable.")
```

```
Coefficient matrix (Lag : 9) is stable.
```

Our VAR model came out to be stable

Step 6: Residual Diagonistics

```
Correlation matrix of residuals
                    GDP Growth  Stock Index  Repo Rate  Unemployment Rate
GDP Growth            1.000000     0.159079  -0.065886          -0.021094
Stock Index           0.159079     1.000000  -0.025062           0.078336
Repo Rate            -0.065886    -0.025062   1.000000          -0.202399
Unemployment Rate    -0.021094     0.078336  -0.202399           1.000000
```

Answers

Step 5:Granger-casuality Test:

```
Granger Causality Test:
GDP Growth and Stock Index:
 p-value = 0.9013235862800828

GDP Growth and Repo Rate:
 p-value = 0.7644950963696011

GDP Growth and Unemployment Rate:
 p-value = 0.6418678840711644

Stock Index and GDP Growth:
 p-value = 0.652750858185938

Stock Index and Repo Rate:
 p-value = 0.17097500847402602

Stock Index and Unemployment Rate:
 p-value = 0.8911552776259037

Repo Rate and GDP Growth:
 p-value = 0.8871624903095431

Repo Rate and Stock Index:
 p-value = 0.8649300548370771

Repo Rate and Unemployment Rate:
 p-value = 0.08667325027565993
```

Answers

```
Repo Rate and Stock Index:
 p-value = 0.8649300548370771

Repo Rate and Unemployment Rate:
 p-value = 0.08667325027565993

Unemployment Rate and GDP Growth:
 p-value = 0.10204711026518772

Unemployment Rate and Stock Index:
 p-value = 0.3307516050186201

Unemployment Rate and Repo Rate:
 p-value = 0.04440428387603912 6
```

Answers

Q2 . Find out "Is there any linkage between GDP and Energy Consumption on Carbon emission and on expenditure on Environment" for given data set (1980-2022) Using required model analysis for Each Country and Compare the Result with Conclusive Remark on Finding.

The Granger causality test results provide insights into whether GDP and Energy Consumption Granger-cause Carbon Emission in your two datasets. Here's a breakdown of how to interpret these results:

For Country A:
GDP -> Carbon Emission:

At lag 1, the F-test and chi-square test p-values are greater than 0.05, indicating that GDP does not Granger-cause Carbon Emission at this lag.
At lag 2, the tests also yield p-values above 0.05, reinforcing that there's no Granger causality from GDP to Carbon Emission at lag 2.
Energy Consumption -> Carbon Emission:

At lag 1, similar to GDP, the p-values are above 0.05, suggesting no Granger causality from Energy Consumption to Carbon Emission at this lag.
At lag 2, the results are consistent, with p-values again above 0.05, indicating no Granger causality at this lag either.
For Country B:
GDP -> Carbon Emission:

At lag 1, the F-test and chi-square test show p-values below 0.05, indicating a Granger causality from GDP to Carbon Emission. This suggests that past values of GDP provide significant information in predicting future values of Carbon Emission at lag 1.
At lag 2, the p-values are above 0.05, suggesting that when considering the past two years, GDP does not Granger-cause Carbon Emission.
Energy Consumption -> Carbon Emission:

At lag 1, p-values are above 0.05, indicating no Granger causality from Energy Consumption to Carbon Emission.
At lag 2, the p-values remain above 0.05, suggesting no Granger causality at this lag as well.

Answers

Summary of Interpretation:
For Country A: There's no evidence of Granger causality from either GDP or Energy Consumption to Carbon Emission at lags 1 and 2.
For Country B: GDP at lag 1 Granger-causes Carbon Emission, indicating a predictive relationship where past GDP values can help forecast future Carbon Emission levels. However, no such causality is detected from Energy Consumption.
These results imply that in Country B, economic activity (as measured by GDP) has a predictive relationship with Carbon Emissions in the short term (lag 1), but this predictive power doesn't extend to Energy Consumption or to longer lag lengths in the context of this analysis. For Country A, the economic and energy consumption variables do not have the same predictive capacity for Carbon Emissions based on the Granger causality tests.
These ARDL model results show the relationship between GDP and Energy Consumption on the Expenditure to Protect the Environment for two datasets, assumed to be Country A and Country B. Here's how to interpret the key elements of these results:

Country A:
Dependent Variable: Expenditure to Protect the Environment.
Model: ARDL(10, 0, 0), indicating up to 10 lags for the dependent variable and no lags for the independent variables.
GDP (Bn).L0: The coefficient is 0.0337, but it's not statistically significant ($p=0.204$), suggesting that current GDP does not have a significant immediate impact on the Expenditure to Protect the Environment.
Energy Consumption.L0: The coefficient is -0.0006, also not significant ($p=0.889$), indicating that current Energy Consumption does not have a significant immediate impact on the Expenditure to Protect the Environment.
Lagged Effects: Various lags of the Expenditure to Protect the Environment show different levels of significance, suggesting that past values have mixed influences on current expenditure levels.
Country B:
GDP (Bn).L0: The coefficient is 0.0258 and is statistically significant ($p < 0.001$), indicating a significant positive impact of current GDP on the Expenditure to Protect the Environment.
Energy Consumption.L0: The coefficient is -0.0040, approaching significance ($p=0.065$), suggesting that current Energy Consumption might have a negative impact on the Expenditure to Protect the Environment, but this is not definitively confirmed at the usual 0.05 significance level.
Lagged Effects: Similar to Country A, the lagged effects show varying influence, with some lags showing potential significance, indicating the historical values of expenditure might influence current levels.

Comparative Insights:
Impact of GDP: GDP has a significant positive impact on the Expenditure to Protect the Environment in Country B but not in Country A. This suggests that economic activity in Country B is more closely tied to environmental protection expenditure.
Impact of Energy Consumption: Energy Consumption does not show a significant immediate impact in either country, but there is a hint of a negative relationship in Country B.
Temporal Dynamics: The significance of various lags in both countries indicates that past expenditures to protect the environment influence current expenditure levels, showcasing the dynamic nature of environmental protection efforts.
Conclusions:
The expenditure to protect the environment in Country B is more responsive to economic changes compared to Country A.
The role of Energy Consumption in determining environmental protection expenditure is less clear and might require further investigation.
The significant lag effects suggest that environmental protection expenditure decisions are influenced by their own past values, indicating potential inertia or delayed responses to changes in GDP and Energy Consumption.